The Data Mine – 3 Minute Thesis

Fredrick National Laboratory

The Cancer-Imaging Archive (TCIA)

The Cancer Imaging Archive (TCIA) is a public resource that stores a large collection of **cancer-related medical images**. These images are carefully **de-identified to remove patient information** and are made **freely available** for researchers and the public to download and use.

Background Information

Data Standardization Script: Data is generated by different researchers with inconsistent formats, making standardization difficult

- Goal: Program a script to map datasets into a common framework to ensure consistent structure and easier integration

CICADAS Data Description: Description is currently treated as a single question limiting section-by-section refinement and feedback.

- Goal: Separate sections and allow for AI suggestions

01

CICADAS Description Wizard

Motivation

CICADAS: Cancer Imaging Checklist for Data Sharing

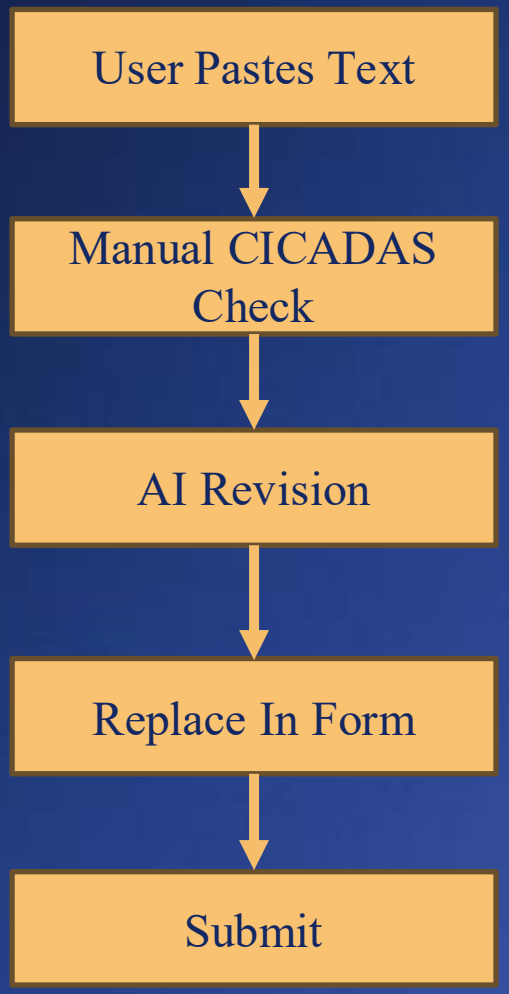
- If a dataset does not include sections in CICADAS, the review process becomes much slower

Solution: Implement an AI wizard which will analyze the dataset to complete any missing sections

- Allows datasets to quickly becomes CICADAS compliant
- Improves descriptions for CICADAS sections

Workflow + Tools

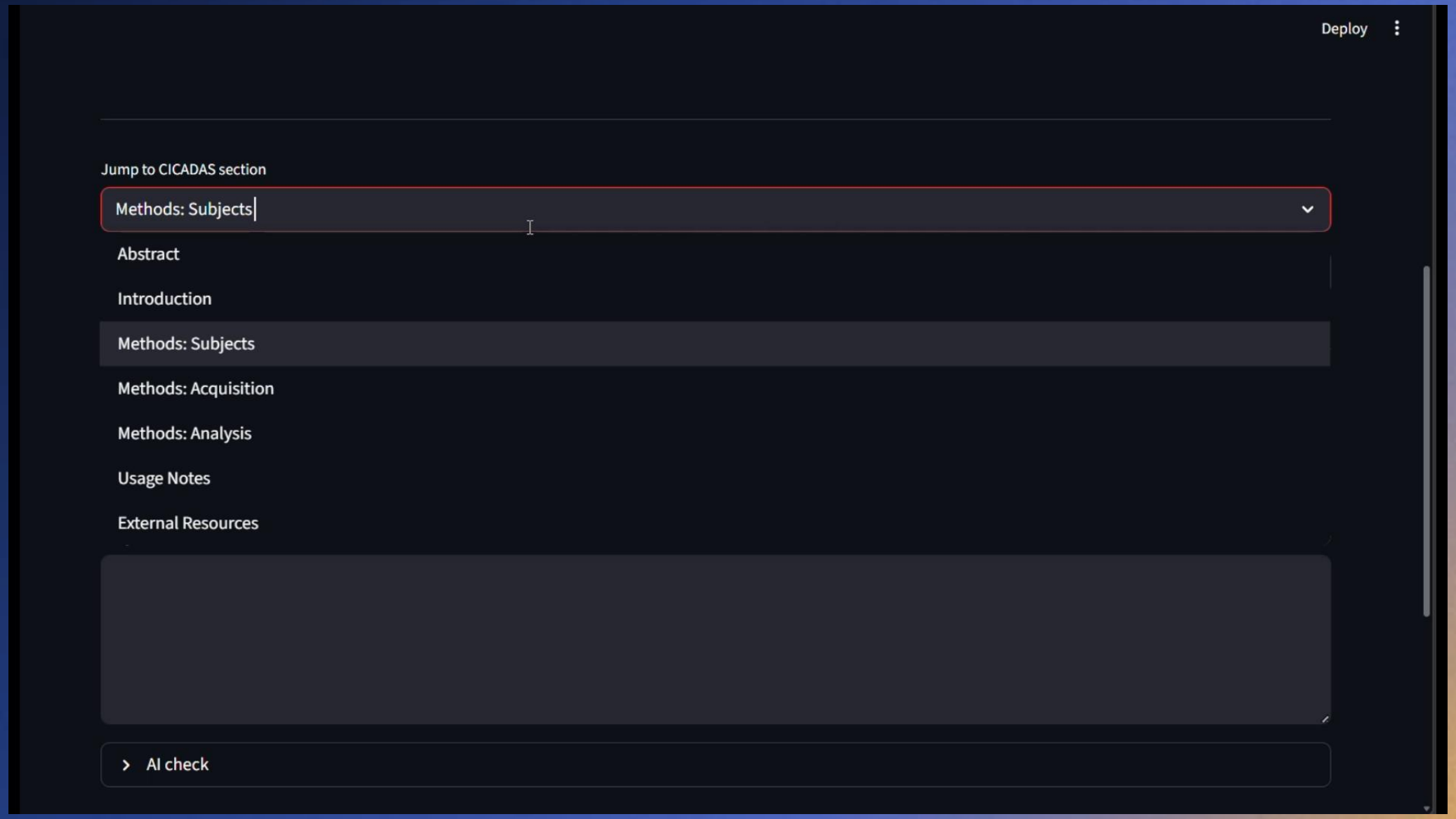
Workflow



Tools

- User Interface: Streamlit
- CICADAS Keyword Detector:
- AI Wizard: Ollama's local Llama 3B model

Demonstration



02

TCIA Data-Standardization Script

Motivation

TCIA datasets contain various imaging files (ex. DICOM, NIfTI) and various clinical spreadsheet. The data is often large, complex, and inconsistently organized.

Solution: Develop a data inventory and metadata extraction script

- Helps pre-populate submission metadata, reducing manual work
- Creates a consistent data inventory

Workflow + Tools

Workflow

Tools

DICOM metadata extractor: pydicom
NIfTI metadata extractor: NiBabel
Analyze spreadsheet: pandas

Demonstration

```
Enter path to data directory (blank = current): C:\Users\jihol\Downloads\dicom test
Create one ZIP per DICOM Series? (y/n): n
Starting data inventory...
Extracting metadata from 702 DICOM files...
  Processed 0/702
  Processed 100/702
  Processed 200/702
  Processed 300/702
  Processed 400/702
  Processed 500/702
  Processed 600/702
  Processed 700/702
✓ Series-level TSV written to dicom_series_inventory.tsv
No NIfTI files found.
Generating Master Data Dictionary template...
  No spreadsheets found to generate a data dictionary.
✓ Inventory JSON written to data_inventory.json

Data Inventory & Standardization complete.

Process finished with exit code 0
```

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References

- The Cancer Imaging Archive. (n.d.). *The Cancer Imaging Archive – A large archive of medical images of cancer.* Retrieved March 10, 2026, from https://www.cancerimagingarchive.net